

configured. These credentials, either unique or user-defined, are shared with the client to connect the device to a network and account, following a setup process similar to that used by commercially available smart bulbs and controllers. Fig. 2 shows

a code snippet that configures the RainMaker device.

Next, the required lights, relays, and sensors are added, and their respective pins are defined. In this implementation, the controller includes a motion sensor, two

lights, one fan, one switch, and an illuminance sensor. Touch pins 1, 3, 4, and 5 are assigned to enable touch-based on or off control for all devices.

A provisioning function is implemented to generate a QR code that enables the device to be added to the app and configured with Wi-Fi credentials. A reset function is also implemented using the boot button on pin 0 of the board. Pressing and holding the boot button for three seconds resets the Wi-Fi configuration, while holding it for ten seconds restores the device to factory settings and restarts the provisioning process. Fig. 3 shows the code snippet illustrating the provisioning function.

Circuit diagram

Fig. 2 shows the system's circuit diagram. It is built around IndusBoard Coin (MOD1), three relay modules (RL1-RL3), two NeoPixel LED displays (L1, L3), and a few additional components.

The three relay modules are used for fan control. If fan speed control is required, the code includes an option for that. The relay can be replaced with a fan speed regulator module that accepts a PWM input signal to control the fan. Alternatively, a custom speed regulator or solid-state relay (SSR) can be designed based on system requirements.

The IndusBoard can power the 3.3V AC-to-DC adaptor. The light and relay can be powered using the Vcc (5V DC). For RGB light PCB modules consuming 9-10 watts, power can be supplied using a 220V AC line. The data pin connects to the I/O pin of the board.

Provisioning the device

First, upload the source code to the IndusBoard. After the upload is complete, open the serial monitor. A link for generating the provisioning QR code will be displayed. Copy this link and paste it into a web browser

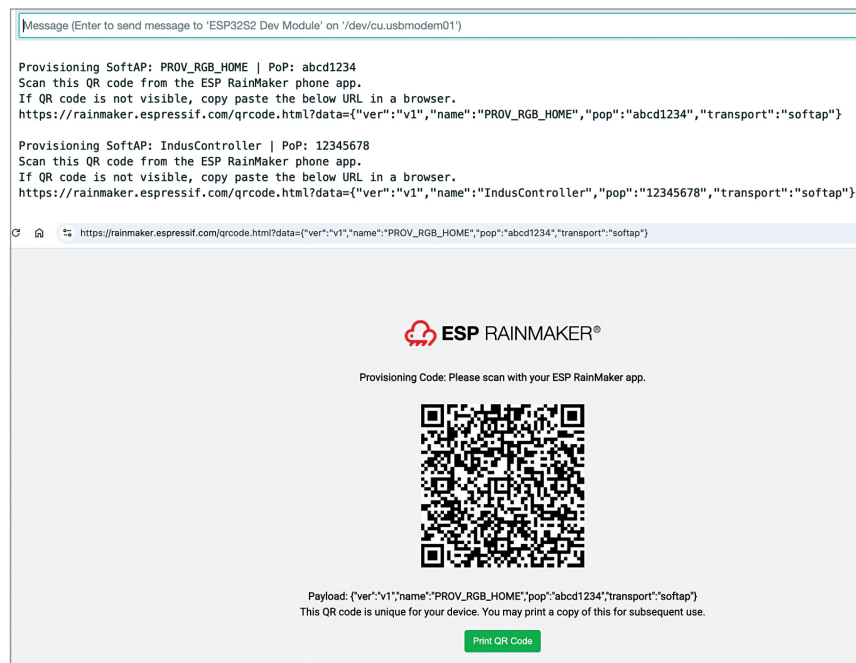


Fig. 5: Device QR code generated for provisioning

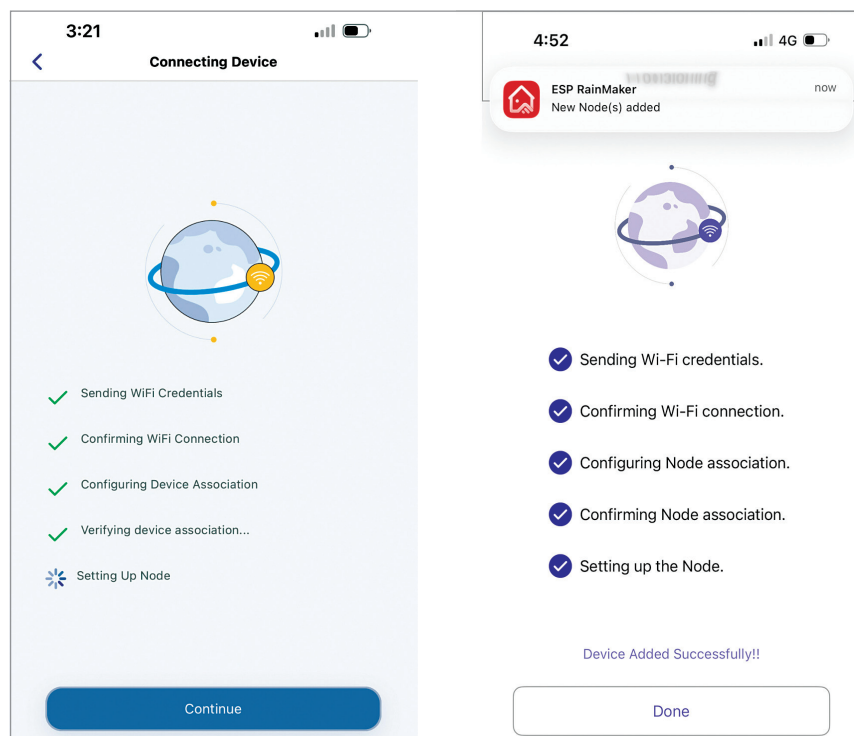


Fig. 6: App provisioning the device and adding it to the RainMaker platform app